

Physical Models

Description

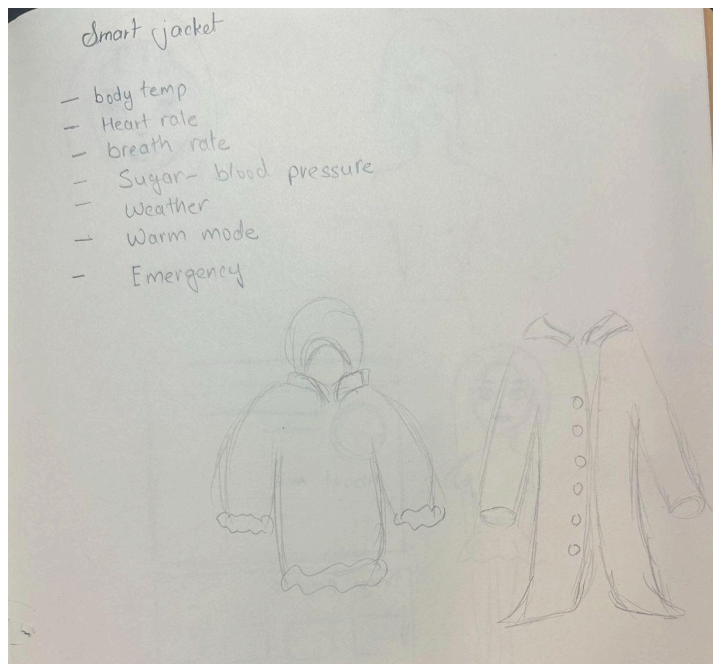
Our project at a basic level is a wearable computing device called the SmartWear Wellness Jacket, designed to help users monitor their health and stay comfortable throughout the day. The jacket uses built-in biometric sensors to track the wearer's heart rate, breathing rate, skin temperature, blood pressure, and environmental conditions such as outside temperature.

One of the main features of the jacket is a smart sleeve display that allows users to quickly view their vitals and weather information directly from the jacket. The sleeve display also allows users to manually adjust the jacket's internal temperature settings based on their comfort level and surrounding weather conditions.

The jacket includes a built-in blood pressure monitoring device that allows users to perform blood pressure tests directly from the jacket and instantly view their results on the sleeve display or companion mobile app. The connected mobile app provides additional features such as health tracking history, personalized notifications, device syncing, and wellness monitoring over time.

Our goal was to create a wearable device that combines health monitoring, convenience, comfort, and adaptive technology into an easy-to-use everyday product.

Early Design Sketches of Prototype



Thin & Stylish

Detachable
blood pressure
attachment



health monitoring apps



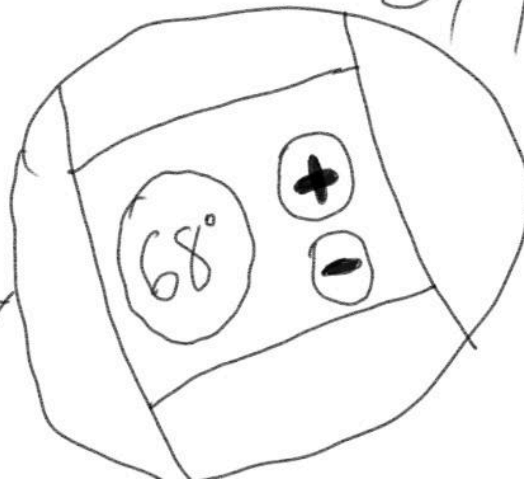
swipe

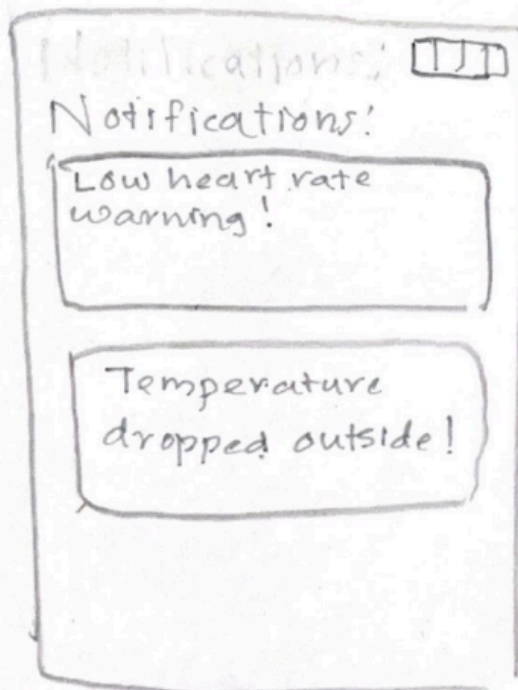


environmental information

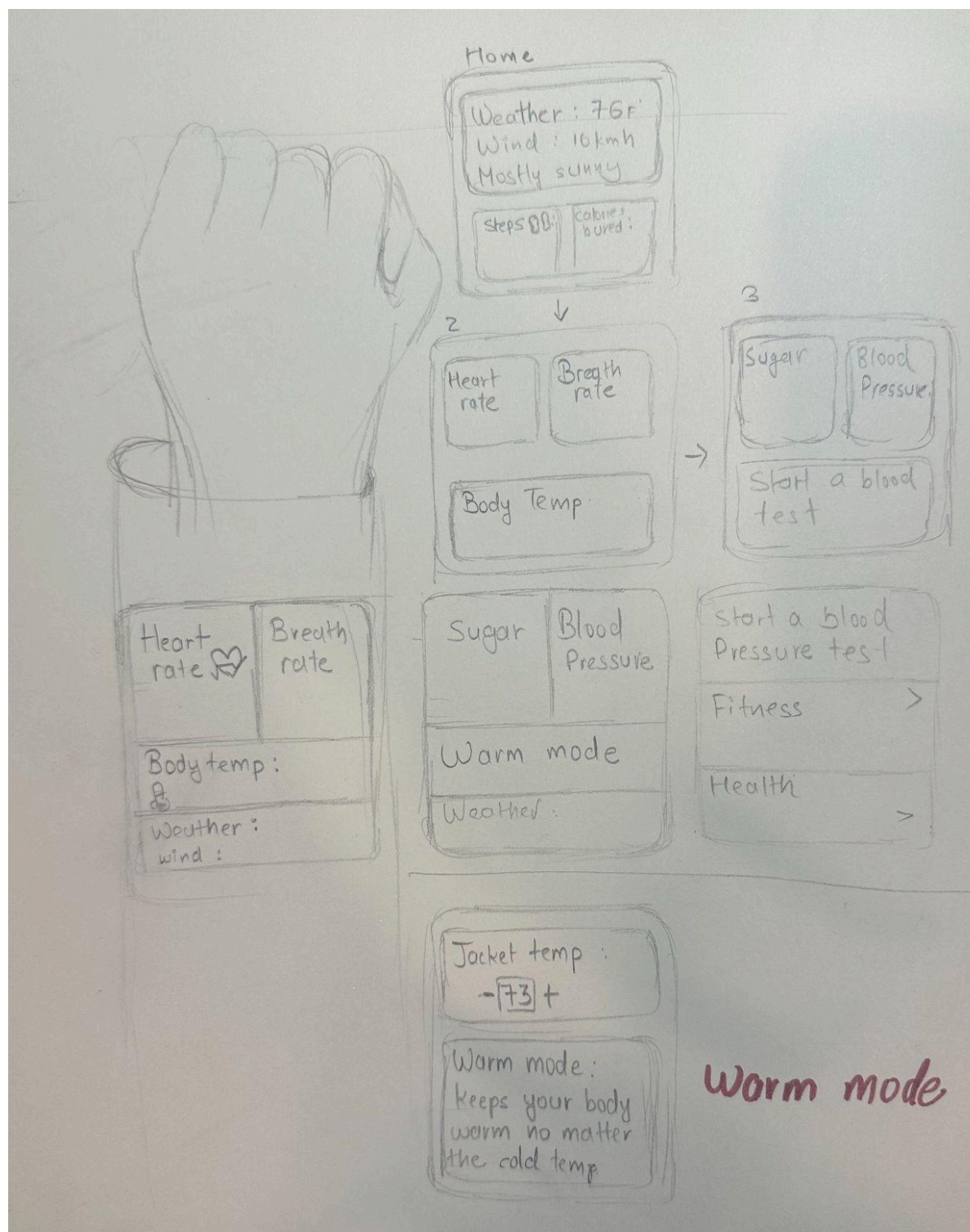


Jacket temperature
monitor





← swipe down to view notifications



Feedback on Sketches

We presented our early sketches to two potential users: a college student who enjoys working out and staying active, and a medical student with knowledge of health and wellness technology. We asked both users for feedback on the overall design, usability, and which features they thought were most useful or unrealistic. We also asked for suggestions on how the product could be improved.

Both users mentioned that the initial sketches were clear and easy to understand. The college student liked that we considered both functionality and style in the jacket design. They suggested offering the jacket in different colors to appeal to a wider range of users. Based on this feedback, we decided to use a slim, unisex jacket with a neutral color for the physical prototype while keeping additional color options as a future improvement.

The college student also emphasized the importance of having the blood pressure device built directly into the jacket rather than detachable. In our original concept, the blood pressure monitor was removable, but the user explained that having the feature integrated into the jacket would be more convenient and useful during workouts or emergencies. Based on this feedback, we redesigned the prototype so the blood pressure device was fully built into the jacket.

The medical student provided feedback about the technical limitations of some of our original ideas. They mentioned that accurately measuring blood sugar levels through a jacket would not be realistic without specialized medical equipment. After considering this feedback, we decided to remove the blood sugar feature from the design to make the product more believable and practical.

The medical student also suggested adding automatic temperature adjustment based on the user's skin temperature and allowing users to switch between Fahrenheit and Celsius. We used this feedback to improve our mid-fidelity prototype by adding a toggle to switch between units of temperature as well as adding both manual and automatic temperature controls. Users can now either adjust the jacket temperature directly from the sleeve display or allow the jacket to automatically respond to body temperature and weather conditions.

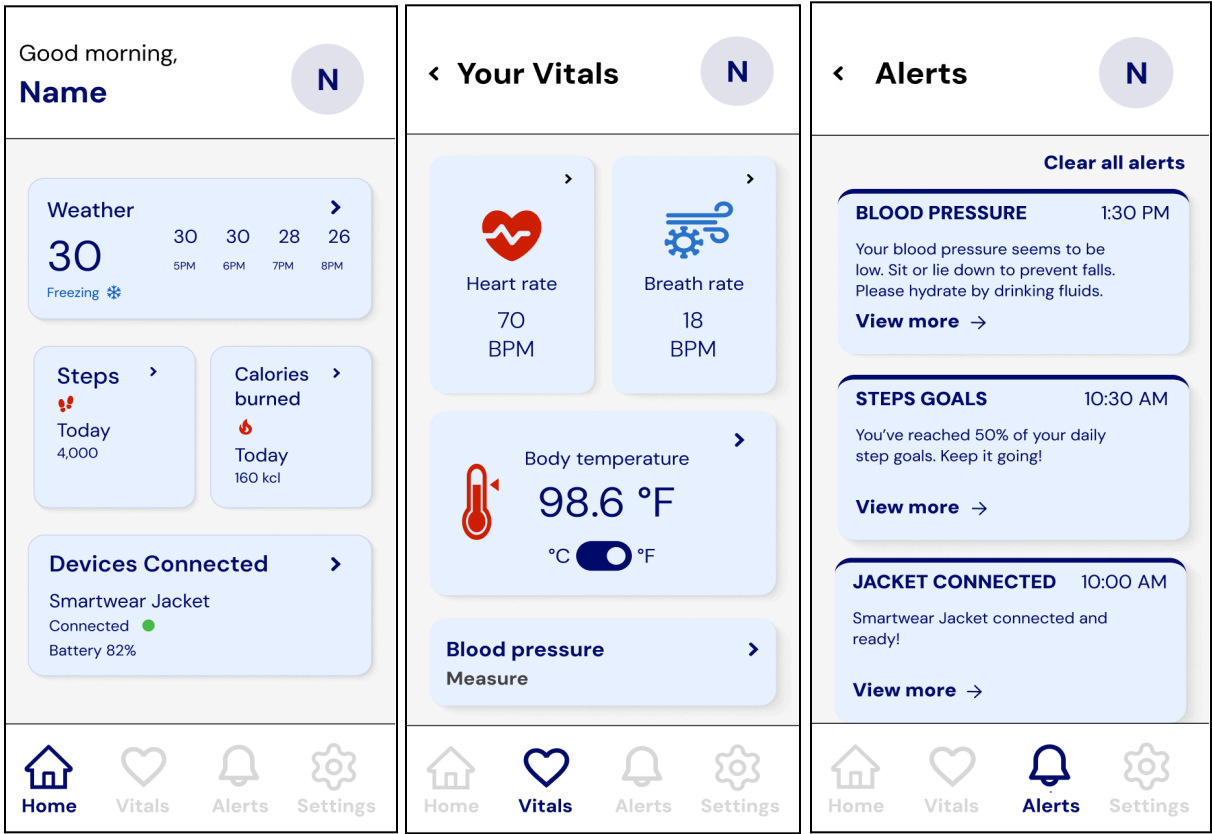
Overall, all the user feedback was very useful as we were able to use their input to refine our product and make improvements. It was impactful hearing valuable feedback from potential users' points of view.

Physical Model Photos

Smart Wear



Mobile App Interface Screens:

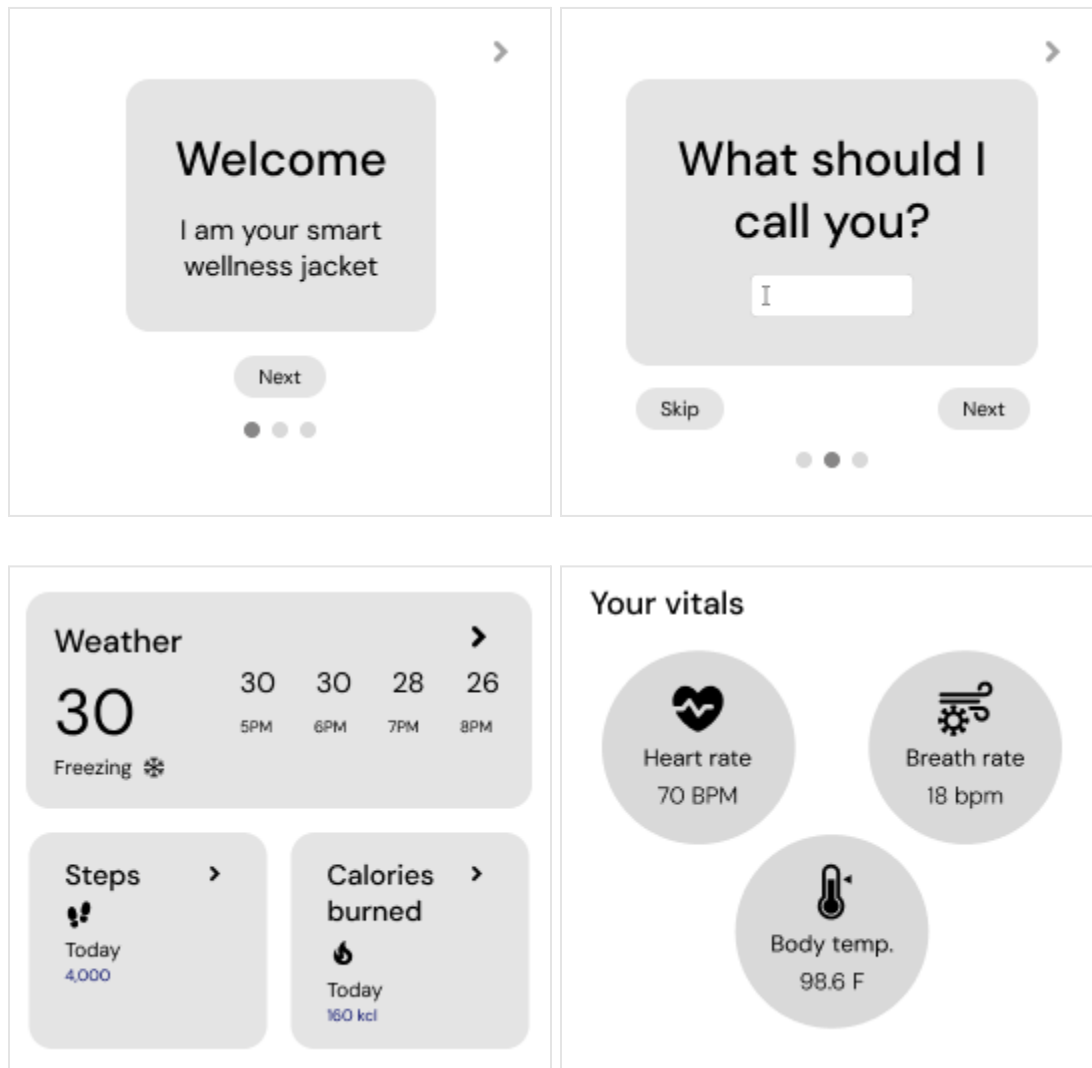


Home Screen

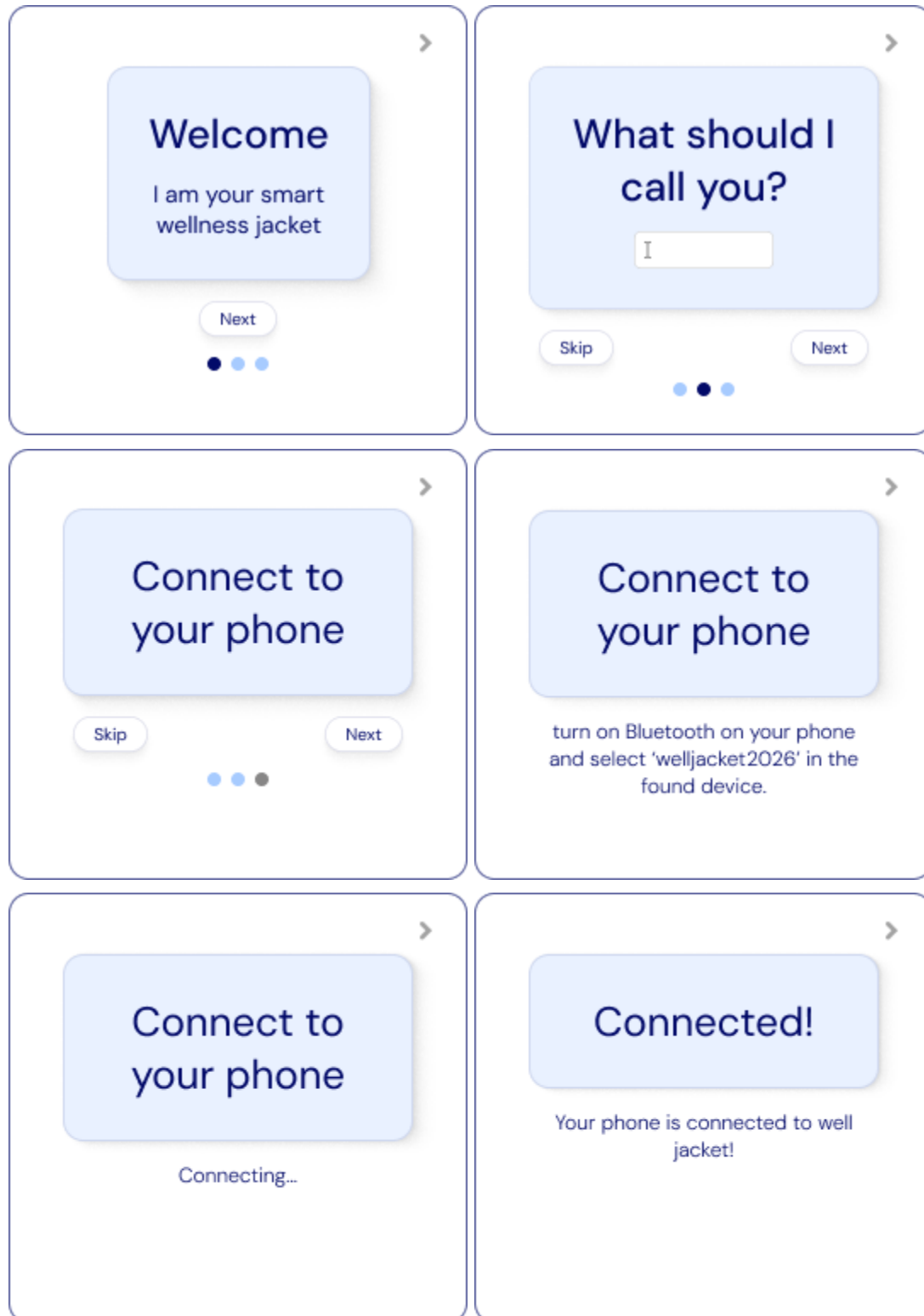
Vitals

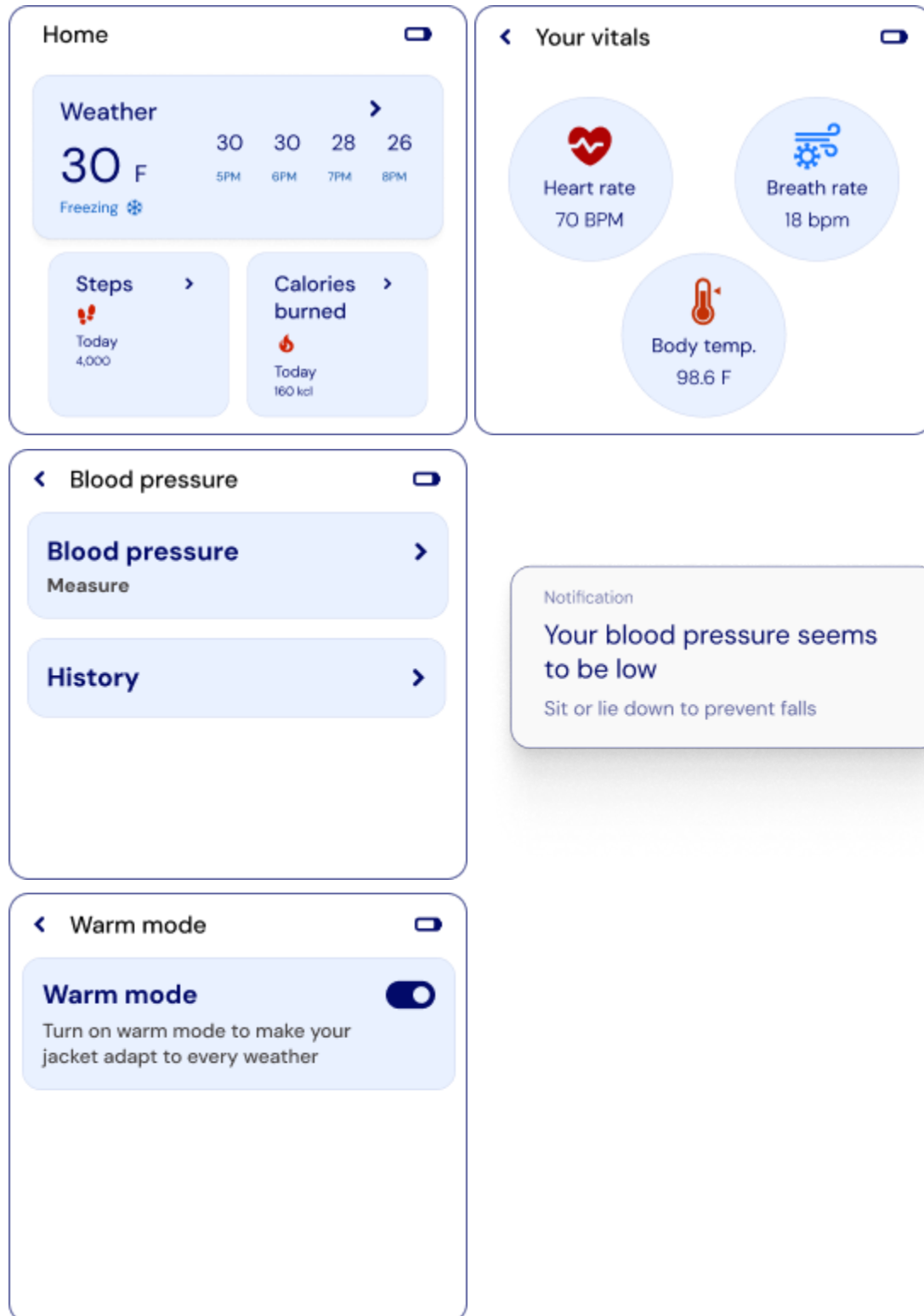
Alerts/Notifications

Mid-fidelity screens- on jacket sleeves



High fidelity screens-on jacket sleeves





Prototype link: [Prototype](#) (swipe or drag for vitals and last screens)

Click to see the [prototype video](#)

Design Justification

The goal of our project was to design a smart jacket that could combine health monitoring, comfort, and convenience into one product. Throughout the design process, we focused on creating a wellness jacket that felt realistic and easy to use. We wanted the SmartWear Wellness Jacket to help users monitor their health in real time while also improving comfort through adaptive technology.

One of the biggest design priorities was making the jacket convenient and accessible for users. Early in the design process, we realized that users would not want to constantly rely on a phone to check important information. Because of this, we added a smart sleeve display directly into the jacket. The sleeve display allows users to quickly check their heart rate, breathing rate, body temperature, blood pressure readings, weather conditions, and alerts without needing to open the mobile app. We designed the interface with large cards, simple layouts, and clear icons so users could easily read information while walking or exercising.

We also designed a companion mobile app to support more detailed features. The app allows users to connect the jacket to their phone, track health history, manage notifications, view blood pressure history, and monitor connected devices and battery life. During the design process, we realized that the sleeve display and the app should serve different purposes. The sleeve display was designed for quick interactions, while the mobile app was meant for more detailed tracking and customization.

User feedback played an important role in refining our design and making the product more realistic and practical. Based on the feedback we received, we decided to fully integrate the blood pressure monitor into the jacket for better convenience and usability. We also removed unrealistic features, such as blood sugar monitoring, and improved the temperature regulation system by adding both manual and automatic controls along with the option to switch between Fahrenheit and Celsius.

The physical prototype, video prototype, and mid-fidelity screen prototypes all helped us refine the project. Creating the physical model helped us think more carefully about comfort and how users would interact with the jacket in real life. The video prototype helped us visualize the user experience step-by-step, while the screen prototypes helped us improve navigation, alerts, and the layout of information on both the sleeve display and mobile app.

Overall, throughout this project we were able to make improvements on our design through the various steps of the design process. If we continued developing the SmartWear Wellness Jacket, we would focus on improving comfort, styling, battery life, sensor accuracy, and adding more customization options for users.